

Sandeep Adhikari

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SUMMARY

Backend software engineer with experience developing automation systems, backend APIs, and distributed service workflows using Python, TypeScript, Node.js, and SQL technologies. Skilled in building modular backend services, integrating third-party APIs, managing asynchronous processing workflows, and debugging reliability issues across production-style environments. Experienced with PostgreSQL, Docker, AWS, and backend infrastructure, with additional project experience in AI evaluation systems and multi-stage workflow orchestration.

EXPERIENCE

Junior Backend Software Engineer

Vertex Systems — Jersey City, NJ

Feb 2024 – Present

- Developed backend APIs in Python and Node.js for internal automation and reporting systems
- Assisted in building document-processing services handling XML and JSON transformation workflows
- Wrote SQL queries and contributed to PostgreSQL schema maintenance
- Supported asynchronous task-processing services for long-running backend operations
- Integrated third-party APIs including Gmail, Slack, and cloud storage providers
- Assisted in debugging authentication, session handling, and API reliability issues
- Contributed to backend monitoring, structured logging, and retry workflows
- Participated in deployment and maintenance workflows for internal backend services

Software Engineering Summer Intern

CloudForge Labs — Newark, NJ

May 2023 – August 2023

- Assisted in developing backend tools for workflow automation and internal data synchronization
- Built REST API endpoints and contributed to TypeScript backend services
- Wrote integration tests and debugging utilities to improve API reliability
- Participated in cloud deployment workflows and infrastructure maintenance
- Collaborated with engineers on feature implementation and issue resolution

TECHNICAL PROJECTS

AI Agent Evaluation Platform

- Built internal evaluation framework for multi-turn coding agents
- Designed milestone-driven evaluation systems with strict continuation criteria
- Implemented binary grading pipelines for observable tool behavior analysis
- Created automated failure-detection logic for hallucinations and unsafe actions

Distributed Document Comparison Engine

- Developed Word-compatible document revision engine generating Track Changes XML directly inside .docx packages
- Built token alignment and diff pipelines using LCS-based comparison systems
- Processed complex document structures including tables, headers, footers, and nested XML content
- Debugged formatting edge cases, revision alignment failures, and whitespace corruption issues

Persistent Workflow Automation System

- Developed AI-assisted workflow engine integrating Gmail APIs, Supabase, PostgreSQL, and external APIs
- Implemented long-running session persistence and execution recovery systems
- Built modular orchestration pipeline separating acquisition, reasoning, execution, and reporting stages

SKILLS

Languages: Python, TypeScript, JavaScript, Go, SQL, C++

Backend: Node.js, Express, FastAPI, REST APIs, WebSockets

Databases: PostgreSQL, MySQL, Redis, Supabase

Cloud & Infrastructure: AWS (EC2, RDS, S3), Docker, CI/CD, Linux

AI & Automation: LLM evaluation, agent workflows, prompt engineering, multi-turn orchestration, tool integrations

Security & Reliability: Prompt injection analysis, access control validation, workflow auditing, structured logging

EDUCATION

New Jersey Institute of Technology — Newark, NJ
B.S. Computer Science